INSTAGRAM

SENTIMENT ANALYSIS

Python | SQL (MySQL) | Power BI

Project Type	End-to-End Data Analytics Project
Domain	Natural Language Processing / Social Media Analytics
Dataset	Instagram Comments (Raw CSV)
Tech Stack	Python (Pandas, NLTK VADER), MySQL, Power BI
Goal	Transform raw user comments into actionable sentiment insights
Date	April 2026

Prepared as a complete Data Analyst Portfolio Project

01 PROJECT OVERVIEW

What Is This Project?

This project delivers a full data analytics pipeline for analyzing Instagram user comments, specifically targeting product-related feedback for a skincare brand (@beminimalist__). The objective is to classify sentiment, extract patterns, and visualize insights in an interactive dashboard.

Goal: Convert raw Instagram comments into business intelligence using NLP, SQL, and dashboards.

Dataset: Real Instagram comments with noise — emojis, URLs, special characters, and mixed languages.

Pipeline: Python cleaning → Sentiment scoring → MySQL storage → Power BI visualization.

Key Metrics at a Glance

780	376	60	344
Total Comments	Positive	Negative	Neutral

Dashboard: Overview Sheet

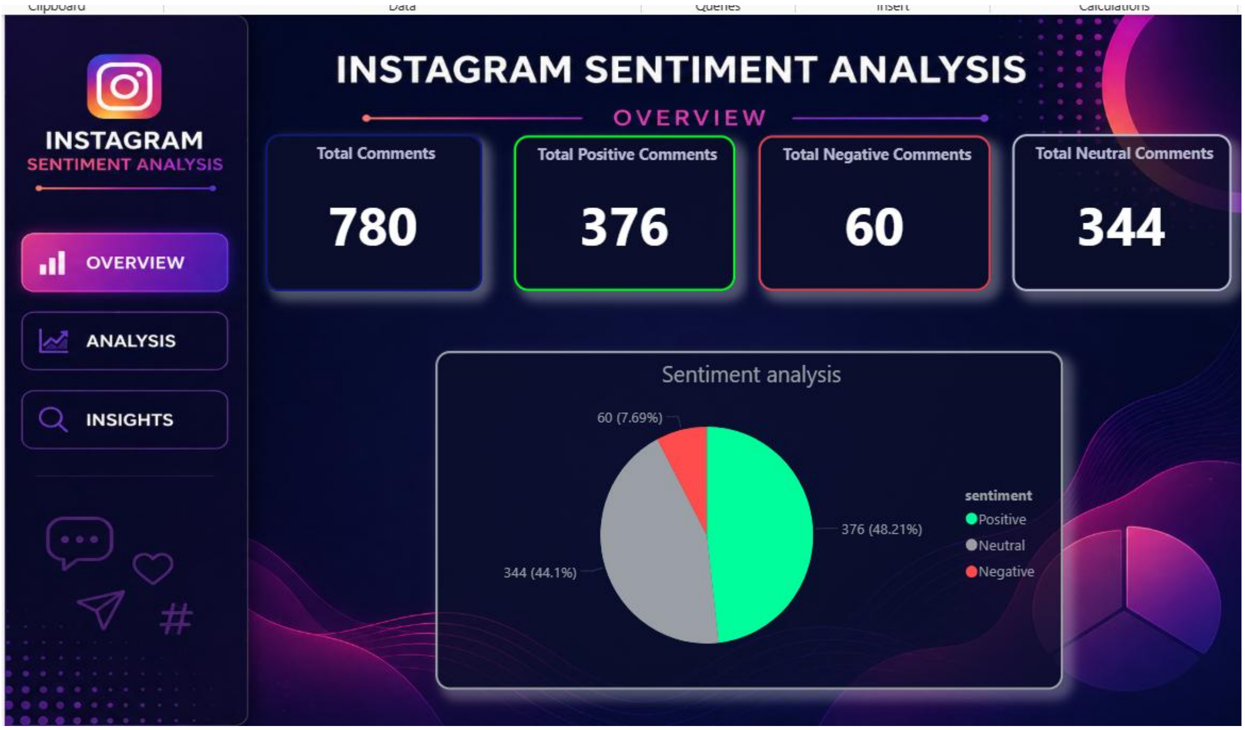

Figure 1: Power BI Overview Dashboard — KPI Cards & Sentiment Pie Chart

02 DATA FLOW & ARCHITECTURE

End-to-End Pipeline

01	**Instagram Dataset** [Raw Comments CSV] Raw user comments scraped or exported from Instagram. Contains noise: emojis, URLs, special characters, mixed scripts.
02	**Python Processing** [Pandas + NLTK VADER] Data cleaning, text normalization, sentiment analysis, and feature engineering. Outputs a structured clean CSV.
03	**Processed Data** [Clean CSV File] Structured dataset with columns: comment, clean_comment, sentiment, comment_length, word_count.
04	**MySQL Database** [SQL Analysis] Data stored in relational DB. SQL queries used for aggregations, distributions, keyword search, and data fixes.
05	**Power BI Dashboard** [3-Sheet Report] Interactive dashboard with KPI cards, pie charts, bar charts, and filtered tables. Three sheets: Overview, Analysis, Insights.

03 PYTHON — DATA PROCESSING & NLP

Step 1: Data Loading & Cleaning

The raw dataset was imported using Pandas. The cleaning pipeline handled all common text noise found in social media data:

- **Load:** Import CSV with Pandas, select only the comment column, drop nulls.
- **Normalize:** Convert to lowercase, remove URLs, special characters, and extra whitespace.
- **Output:** clean_comment — ready for NLP analysis.

Step 2: Sentiment Analysis — VADER

VADER (Valence Aware Dictionary and sentiment Reasoner) from NLTK was used for rule-based sentiment classification:

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
sia = SentimentIntensityAnalyzer()

def classify_sentiment(text):
    score = sia.polarity_scores(text)['compound']
    if score > 0:   return 'Positive'
    elif score < 0: return 'Negative'
    else:           return 'Neutral'

df['sentiment'] = df['clean_comment'].apply(classify_sentiment)
```

Step 3: Feature Engineering

Additional computed columns were added to support quantitative analysis in SQL and Power BI:

```
df['comment_length'] = df['clean_comment'].apply(len)
df['word_count']      = df['clean_comment'].apply(lambda x: len(x.split()))

# Final export for SQL import
df[['comment','clean_comment','sentiment','comment_length','word_count']] \
  .to_csv('instacomments_clean.csv', index=False)
```

04 SQL — DATA STORAGE & ANALYSIS

Database Schema

```
CREATE DATABASE insta_comments;
USE insta_comments;

CREATE TABLE comments (
    id             INT AUTO_INCREMENT PRIMARY KEY,
    comment        TEXT,
    clean_comment  TEXT,
    sentiment      VARCHAR(20),
    comment_length INT,
    word_count     INT
);
```

Key SQL Queries Performed

1. Sentiment Distribution

```
SELECT sentiment, COUNT(*) AS total_comments
FROM comments
GROUP BY sentiment;
```

2. Percentage Distribution

```
SELECT sentiment,
       COUNT(*) * 100.0 / (SELECT COUNT(*) FROM comments) AS percentage
FROM comments
GROUP BY sentiment;
```

3. Average Comment Length by Sentiment

```
SELECT sentiment, AVG(comment_length) AS avg_length
FROM comments
GROUP BY sentiment;
```

4. Keyword Search (Negative Comment Analysis)

```
SELECT *
FROM comments
WHERE clean_comment LIKE '%delay%'
   OR clean_comment LIKE '%problem%'
   OR clean_comment LIKE '%not%';
```

05 POWER BI — DASHBOARD & VISUALIZATION

Sheet 1: Overview

Four KPI cards display total, positive, negative, and neutral comment counts at a glance, color-coded with neon accents on a dark (#0B0F2A) background. A pie chart visualizes the sentiment split.

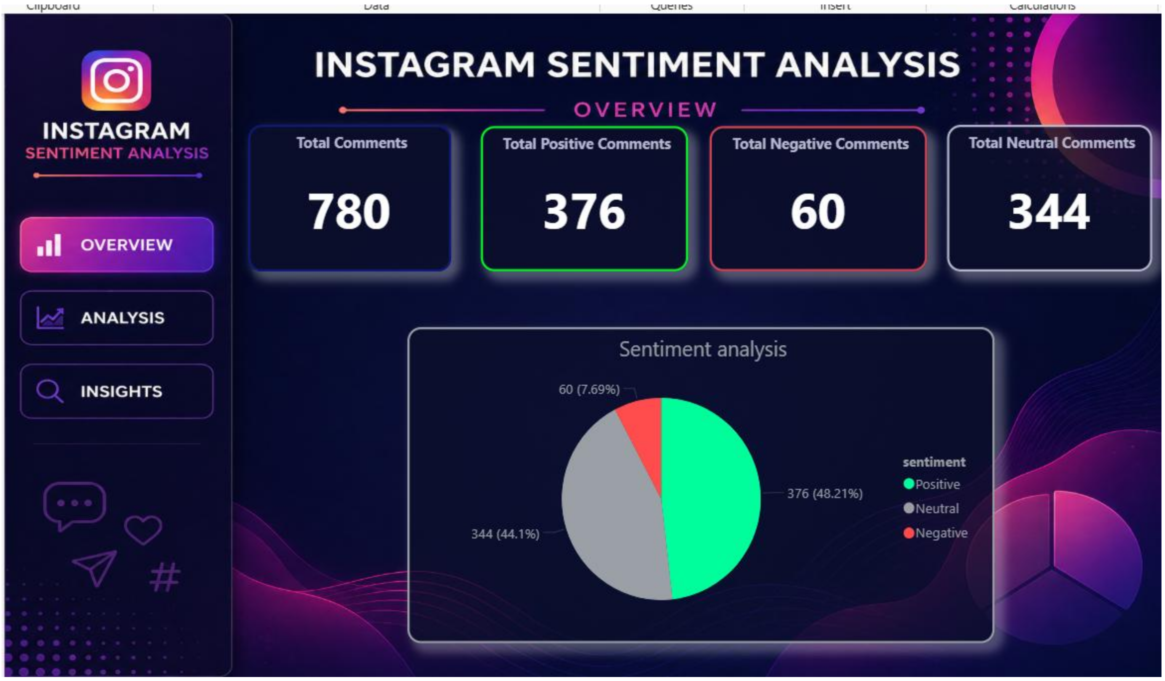

Figure 2: Power BI Overview — KPI Cards & Sentiment Distribution Pie Chart

Sheet 2: Analysis

Two bar charts display average comment length and average word count by sentiment. A detailed table shows individual comments with metrics.

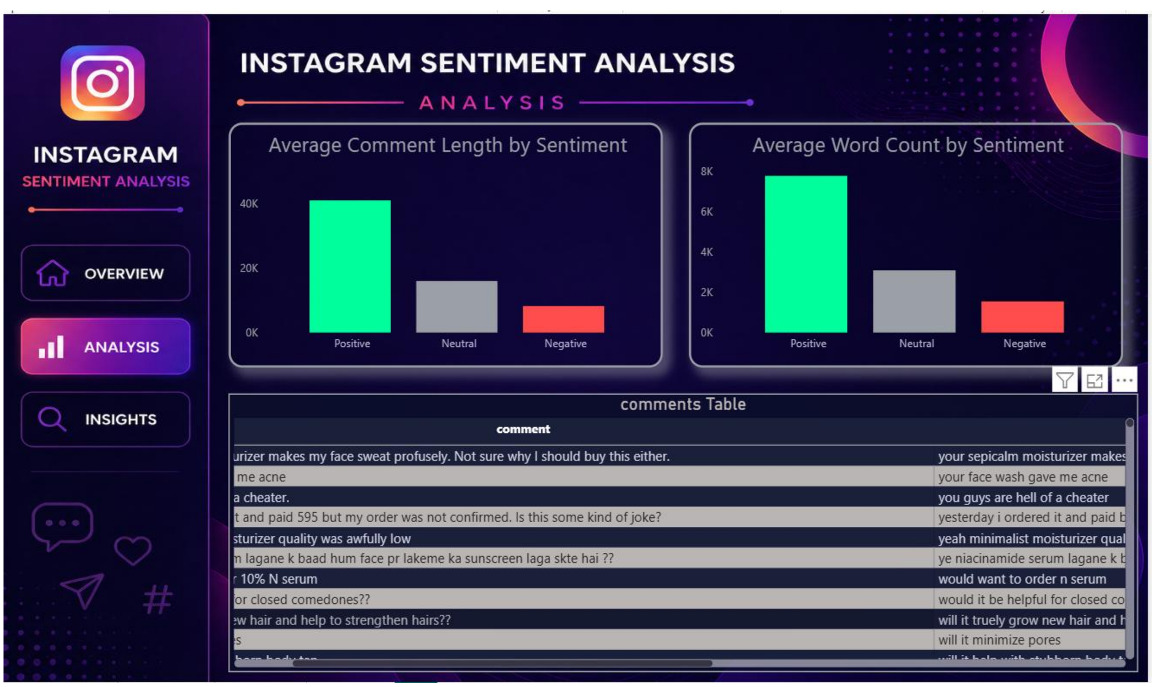

Figure 3: Analysis Sheet — Average Comment Length & Word Count by Sentiment

Sheet 3: Insights

Side-by-side filtered tables display negative vs positive comments for quick comparison. Helps product teams identify pain points and praise simultaneously.

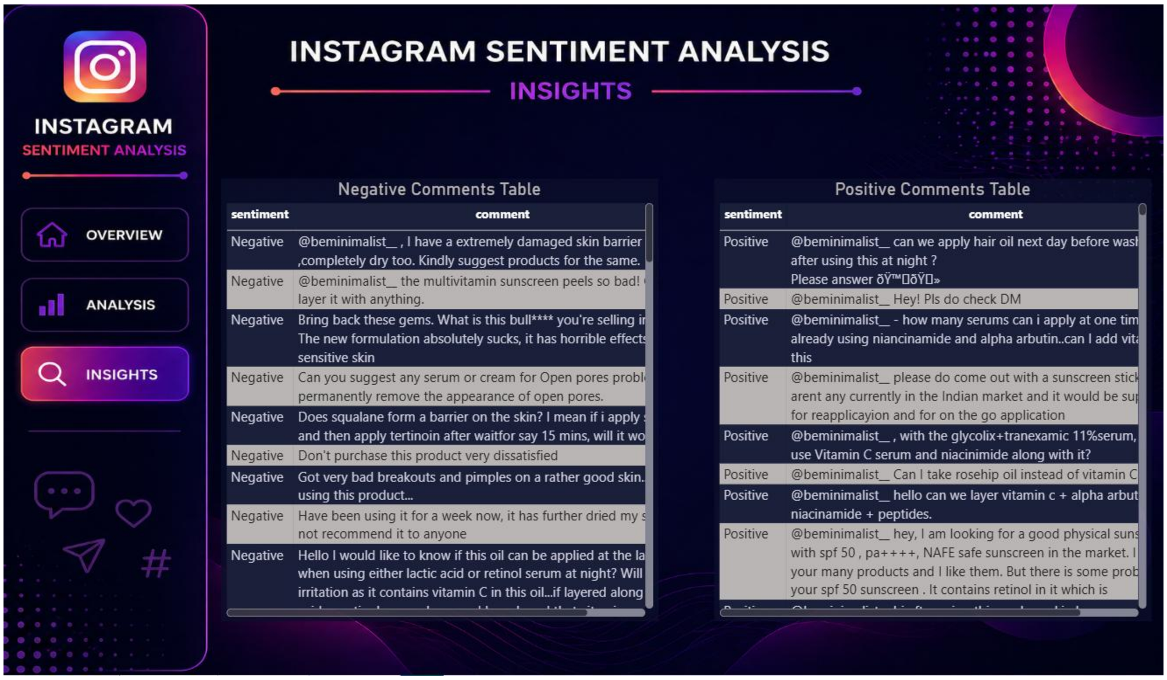

Figure 4: Insights Sheet — Negative vs Positive Comment Tables

06 KEY INSIGHTS & FINDINGS

Data-Driven Observations

- 48.2% of all comments are Positive — indicating strong brand loyalty and customer satisfaction for the skincare brand.
- Negative comments (7.7%) are significantly longer and more detailed, reflecting genuine complaints about product formulation, delivery, or skin reactions.
- Neutral comments (44.1%) represent queries and informational requests — indicating high customer engagement and pre-purchase interest.
- High-frequency keywords include serum, skin, acne, niacinamide, vitamin C — revealing that discussions are product-specific and ingredient-focused.
- Average comment length is highest for positive sentiment (Positive > Neutral > Negative in total length), but negative comments show more words per unique comment.

Business Recommendations

- Address recurring complaints about formulation changes and product reactions through revised quality control and clearer labeling.
- Capitalize on positive comments by using them as testimonials; identify top-praised products for marketing campaigns.
- Automate this pipeline with scheduled scraping to monitor sentiment trends over time and respond faster to negative spikes.
- Expand keyword analysis to include Hinglish terms to capture a broader Indian audience sentiment (e.g., baad, chahiye, lagana).

07 TECHNICAL LEARNINGS

Skills Applied	Tools Mastered
✓ End-to-end pipeline development	⚙ Python (Pandas, NLTK)
✓ Text preprocessing for NLP	⚙ MySQL & SQL querying
✓ VADER sentiment classification	⚙ Power BI DAX & visuals
✓ SQL-based analytical thinking	⚙ Data storytelling
✓ Handling real-world noisy data	⚙ Dark-theme dashboard design

08 PROJECT VALUE SUMMARY

Python
Data cleaning + NLP using real-world noisy data

SQL
MySQL querying, aggregations, and data repair

Power BI
3-sheet interactive dark-theme dashboard

This is a complete, end-to-end Data Analyst portfolio project demonstrating real-world skills across the full data pipeline.